

Revision Notes for Class 12 Chemistry (Chemistry – II)

Chapter 8 – Aldehydes, Ketones and Carboxylic Acids

1. Carbonyl Compounds: Aldehydes, ketones, carboxylic acids, and their derivatives are organic compounds containing the carbonyl group (CO) as the functional group, collectively known as carbonyl compounds.

2. Nature of the Carbonyl Group: The oxygen atom in the carbonyl group is significantly more electronegative than the carbon atom. This causes the electron cloud of the π -bond to shift towards the oxygen, making the carbonyl group polar. As a result, the carbonyl carbon acquires a positive charge, and the carbonyl oxygen carries a negative charge.

3. Methods of Preparation of Aldehydes and Ketones:

Controlled Oxidation: Aldehydes and ketones are produced by the controlled oxidation of primary and secondary alcohols, respectively.

Dehydrogenation of Alcohols: Primary alcohols yield aldehydes, while secondary alcohols produce ketones upon dehydrogenation.

4. Preparation of Aldehydes:

Rosenmund Reduction: Acyl chlorides are hydrogenated using palladium on barium sulfate, partially poisoned with sulfur or quinoline, to prepare aldehydes.

Stephen's Reduction: Nitriles are reduced to imines with stannous chloride in the presence of hydrochloric acid, followed by hydrolysis to give the corresponding aldehydes.

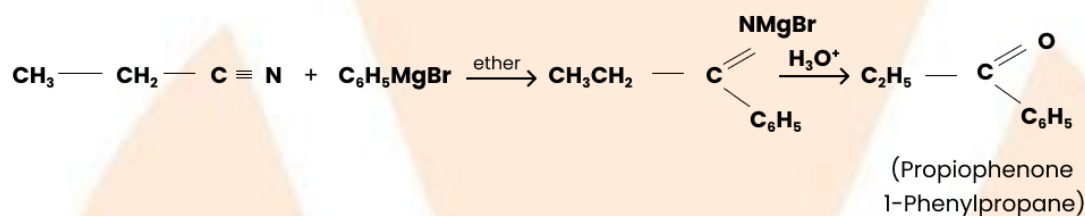
Etard Reaction: Chromyl chloride oxidizes the methyl group of toluene to form a chromium complex, which on hydrolysis yields benzaldehyde.

Gatterman-Koch Reaction: Benzene or its derivatives react with CO and HCl in the presence of anhydrous $AlCl_3$ or CuCl to produce benzaldehyde or substituted benzaldehydes.

5. Preparation of Ketones:

Acyl Chlorides: Treatment of acyl chlorides with dialkyl cadmium, prepared from cadmium chloride and Grignard reagent, produces ketones.

From Ketones: Benzene or substituted benzene reacts with acid chloride in the presence of anhydrous AlCl_3 to form ketones.



6. Properties of Aldehydes and Ketones:

Reactivity: Aldehydes are more reactive than ketones in nucleophilic addition reactions.

Nucleophilic Addition: Both aldehydes and ketones undergo nucleophilic addition reactions with nucleophiles like HCN , NaHSO_3 , alcohols, ammonia derivatives, and Grignard reagents.

Reduction: Aldehydes and ketones reduce to primary and secondary alcohols, respectively.

Clemmensen and Wolff-Kishner Reductions: The carbonyl group is reduced to a CH_2 group by treatment with zinc amalgam and hydrochloric acid (Clemmensen reduction) or with hydrazine followed by heating with NaOH or KOH in a high-boiling solvent (Wolff-Kishner reduction).

Tollen's Test: Aldehydes are oxidized by Tollen's reagent, forming a silver mirror on the test tube, while ketones do not react.

Fehling's Test: Aldehydes reduce Fehling's solution to form a red precipitate of cuprous oxide, but ketones do not give this reaction.

Aldol Condensation: Aldehydes and ketones with at least one α -hydrogen atom undergo condensation with dilute alkali to form β -hydroxy aldehydes or β -hydroxy ketones.

Cross Aldol Condensation: A mixture of two different aldehydes or ketones with an α -hydrogen atom condenses in the presence of dilute alkali to produce a mixture of four products.

7. Cannizzaro Reaction: Aldehydes lacking an α -hydrogen atom undergo self-oxidation and reduction (disproportionation) when treated with concentrated alkali, producing alcohol and carboxylic acid salt.

8. Electrophilic Substitution Reactions: The carbonyl group acts as a deactivating and meta-directing group in electrophilic substitution reactions on the aromatic ring.

9. Methods of Preparation of Carboxylic Acids:

Oxidation: Carboxylic acids can be prepared by the oxidation of primary alcohols and aldehydes.

Side Chain Oxidation: Aromatic carboxylic acids are obtained by oxidizing the side chains of alkyl benzenes.

Hydrolysis: Nitriles and amides can be hydrolyzed to form carboxylic acids.

Grignard Reagent: Carboxylic acids can also be prepared by reacting Grignard reagents with carbon dioxide.

10. Properties of Carboxylic Acids:

Solubility: Aliphatic carboxylic acids with up to four carbon atoms are miscible in water due to hydrogen bonding, but solubility decreases with increasing carbon atoms.

Acidity: Electron-withdrawing groups stabilize the carboxylate anion, increasing acidic strength, while electron-donating groups destabilize it, making the acid less acidic.

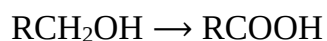
Halogenation: Carboxylic acids with an α -hydrogen atom undergo halogenation at the α -position when treated with chlorine or bromine in the presence of red phosphorus, forming α -chloro or α -bromo carboxylic acids (Hell-Volhard Zelinsky Reaction).

Carboxylic Acids

Carboxylic acids are commonly named by adding the suffix -ic acid. In IUPAC nomenclature the -e is replaced by -oic acid.

Preparations:

1. On oxidising primary alcohols and aldehydes with common oxidising agents such as potassium permanganate or potassium dichromate, we readily obtain carboxylic acids.



2. On oxidation of alkylbenzenes and further hydrolysis of the intermediate formed, we obtain carboxylic acids.

3. Nitriles are hydrolysed to corresponding amides and on further reacting with H^+ or OH^- we obtain carboxylic acids. Amides readily give carboxylic acids on hydrolysis.

4. Grignard reagents on reacting with dry ice in the presence of ether give carboxylic acids salts which on hydrolysis gives us carboxylic acids

5. Hydrolysis of esters in the presence of an acid readily gives carboxylic acids.